SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes **Total Marks: 25**

Code of Conduct (192525447)

I S. Tejayadav (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Tejayadav

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty **Total Marks Awarded:**

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1) Standalone computing refers to computers that operate independently using local hardware, software and storage. As technology evolved, computing shifted to network systems, client-server architecture, distributed computing, virtualization and finally cloud computing.

Key technology transitions:

- Standalone computing: Applications and data stored on a single computer.
- Client-Server computing: clients access servers from centralized servers.
- Distributed computing: Multiple computers share processing.
- Virtualization: Multiple virtual machines run on a single physical server.
- Cloud computing: computing resources are delivered over the internet on demand.

Benefits of cloud computing.

- Scalability
- Cost efficiency
- High availability
- Remote accessibility.

2) A hypervisor is software that creates and manages virtual machines (VMs). It allows multiple operating systems to run on one physical computer by sharing hardware resources such as CPU, memory and storage.

Type 1 Hypervisor	Type 2 Hypervisor
Runs directly on hardware	Runs on top of a host.
Higher performance	Slightly slower.
More secure	Less secure
Used in data centres	used for personal computers
Ex: VMware, ESXi	Ex: Oracle, VMware workstation.

3) An API (Application programming interface) is a set of rules that allows different software applications to communicate.

Role of APIs in Cloud computing:

- Connect different cloud services.
- Enable application integration.
- Automate cloud operations.
- Allow secure data exchange.
- Support third-party app development.

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ating systems.

iteroperability:

APIs enable different cloud platforms and applications to exchange data and work together regardless of programming language or operating system.

Ex: A mobile app uses an API to store files in google Drive.

4) Resource pooling means cloud providers combine computing resources into a shared pool that serves multiple users.

Benefits:

- Efficient utilization of resources.
- Lower infrastructure cost.
- Better performance.
- Resources are allocated dynamically based on demand.
- Supports multiple users simultaneously without affecting service quality.

Ex: Many companies uses the same AWS data centre while each company's data remains isolated and secure.

5) Rapidly elasticity is ability of cloud systems to automatically increase or decrease computing resources based on workload demand.

How it supports dynamic workloads.

- Adds servers during high traffic.
- Removes unused resources during low demand.
- prevents application downtime.
- Improves performance.
- Reduces operational costs by paying only for resources used.

Ex:- During an online shopping festival, cloud servers automatically scale up to handle millions of users and scale down after the sale ends.